SkillUp

Resources: Introduction, Data Handling, and Cleansing

Objective

This reading puts together information and resources that will help you on the job.

Introduction

Data handling and cleansing are fundamental stages in the business intelligence (BI) lifecycle. Data is often collected from multiple sources, such as transactional databases, social media, sensors, and more. This data is typically heterogeneous, voluminous, and prone to errors. Poor data quality can lead to incorrect insights and poor decision making.

Data cleansing, however, focuses on improving data quality by detecting and correcting (or removing) errors and inconsistencies. With the advent of generative AI, these processes have become more sophisticated, efficient, and accurate, transforming how organizations leverage data for decision-making.

Data handling

Data handling involves data integration, data storage, and data access. Let's see how generative AI plays a role in streamlining these processes.

Data integration

Data handling begins with data integration. In data integration, the data from various sources is consolidated into a single, unified view. This process involves extract, transform, load (ETL) processes and data lakes.

Extract, transform, load (ETL): ETL tools extract data from different sources, transform it into a suitable format, and load it into a data warehouse or BI system. Generative AI can enhance ETL processes by automating the transformation steps, such as data mapping and normalization.

Data lakes: Data lakes store raw data in its native format until it is needed. AI-powered tools can index and catalog data in data lakes, making it easier to search and retrieve relevant information.

Data storage

Once integrated, data must be stored efficiently. Generative AI can optimize data storage by predicting data usage patterns and suggesting the most efficient storage solutions, whether on-premises or in the cloud.

Data access

Efficient data handling ensures that users can access the data they need, when they need it. AI can enhance data access by enabling natural language querying, allowing users to interact with data using everyday language. For example, tools like OpenAI's Codex can translate natural language queries into SQL commands, making data access more user friendly.

Data cleansing

Data cleansing, or data scrubbing, involves identifying and rectifying errors in the data set. This process is crucial for maintaining data quality and ensuring reliable BI insights.

Some common data quality issues include:

- **Missing data:** Incomplete records that lack essential information
- **Duplicate data:** Multiple records for the same entity
- **Inconsistent data:** Data that is formatted differently or contains discrepancies
- **Outliers:** Data points that deviate significantly from the rest of the data set

Generative AI for data cleansing

Generative AI can automate many data cleansing tasks, significantly reducing the time and effort required. Some applications include:

Imputation: AI models can predict and fill in missing values based on patterns in the data. For example, generative models like generative adversarial networks (GANs) can be used to generate plausible values for missing data points.

De-duplication: AI algorithms can identify and merge duplicate records by analyzing similarities and patterns, ensuring that each entity is represented uniquely.

Normalization: AI tools can standardize data formats, such as dates and addresses, to ensure consistency across the data set.

Anomaly detection: AI can identify outliers and unusual patterns that may indicate data entry errors or fraudulent activities. For instance, clustering algorithms can group similar data points and highlight those that fall outside the expected range.

AI tools for data handling and cleansing

Several AI tools can enhance data handling and cleansing processes:

Talend: Provides an AI-powered platform for data integration and quality. Features include data profiling and cleansing. You can access the website here.

Alteryx: Combines data integration, cleansing, and advanced analytics, enabling users to prepare data quickly and efficiently. You can access the website here.

Synthetic data

Synthetic data generation plays a significant role in the data handling process. It addresses the following issues.

- **Data scarcity:** New or niche businesses lacking sufficient data required to obtain statistically significant results.
- **Data quality issues:** Inconsistencies and missing values in real-world data
- **Data privacy:** Sensitive data covered under data privacy laws that cannot be shared

AI tools for synthetic data generation

Here are some generative AI tools for synthetic data generation.

Mostly AI: A tool that creates fully anonymous data with the original data. You can access the website here.

Hazy: This tool helps businesses generate synthetic data that can be safely used for testing, analytics, and AI training without risking sensitive information. You can access the website here.

Gretel.ai: A synthetic data platform that creates synthetic versions of existing data in minutes. It can anonymize and enrich data, enabling businesses to generate valuable insights while ensuring privacy. You can access the website here.

Summary

This reading revisited the concepts of data handling and cleansing, and the role of synthetic data in the process. It also provided a list of tools which you can leverage to streamline these activities.